

11. A hydrated compound $M_aAl_bZ_c \cdot dH_2O$ contains an s-block metal ion, M, and a halide ion, Z. The following tests were undertaken.

Test	Result
Heating to constant mass	A 0.0200 mol sample lost 1.44 g when heated
Adding concentrated sulfuric acid	Observations included coloured fumes, an orange-brown solution, steamy fumes and a choking gas; there was no smell of rotten eggs
Adding excess silver nitrate to a solution	A 25.0 cm ³ sample of a solution of concentration 0.203 mol dm ⁻³ produced a precipitate with a dry mass of 3.814 g
Adding excess sodium hydroxide to a solution	A white precipitate formed which dissolved in excess sodium hydroxide
Elemental analysis	The hydrated compound was found to contain 1.63 % by mass of s-block metal and 6.34 % by mass of aluminium

- (a) Calculate the value of d in the formula $M_aAl_bZ_c \cdot dH_2O$. [2]

$d =$

- (b) (i) Identify the halide present in the compound, explaining fully all observations that led to your conclusion. [2]

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(ii) Calculate the value of c in the formula $M_aAl_bZ_c \cdot dH_2O$.

[2]

$c =$

(c) State what the observations with sodium hydroxide indicate about the acidity/basicity of the aluminium ions. [1]

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(d) The M_r of the original **hydrated** compound is 425.62. Use all the information above to find the formula of the hydrated compound. Explain your reasoning. [3]

Formula of the **hydrated** compound

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